

**Forecast accuracy came out
suspiciously high**

**Two LEFT JOINS fixed
every missing month**

ATLIQ HARDWARE GDB0041

MYSQL | Vishnu Ram Sachin | Data Analyst



Problem Statement

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- Yesterday : Fixed running total that broke at the fiscal year boundary
- Today Leadership wants an honest forecast vs actuals comparison per product per month

The Trap ❌

- INNER JOIN on date , customer_code and product_code looks clean but it only keeps month present on both sides

Forecast , Zero sold , Dropped

Sold , Never forecast , Dropped



Requirment

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Keeps every forecasted month ,
even with Zero sales



Keep every sold month ,
even with Zero forecast



Two LEFT JOINs , one filter
not a single INNER JOIN

The Fix

```
FROM fact_sales_monthly fsm
LEFT JOIN fact_forecast_monthly ffm
  UNION ALL
FROM fact_forecast_monthly ffm
LEFT JOIN fact_sales_monthly fsm
WHERE fsm.product_code IS NULL
```



A Quick Analogy

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A Party's Guest list vs It's Sign in sheet

Invited but didn't come

We expected them and sent the invite. They never showed up
(forecast LEFT JOIN, WHERE sale IS NULL)

Not invited , Came anyway

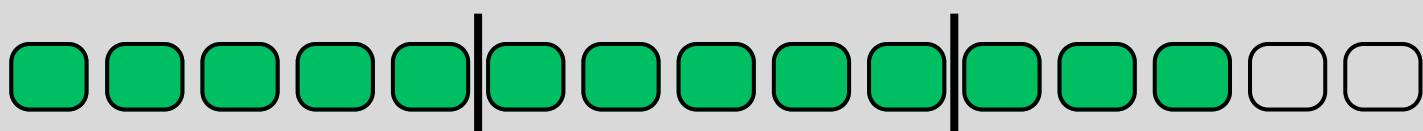
We never invited them , but they showed up and signed in
(sales LEFT JOIN , forecast)

The Full Picture

Both grouped combined with everyone who matched ,
Nobody counted twice
(UNION ALL)



```
WITH forecast_actual_compare_report AS (  
    SELECT  
        fsm.date,  
        MONTHNAME(fsm.date) AS month_name , fsm.customer_code,  
        fsm.product_code,  
        fsm.fiscal_year,  
        COALESCE(ffm.forecast_quantity, 0) AS forecast_quantity,  
        fsm.sold_quantity  
    FROM fact_sales_monthly AS fsm  
    LEFT JOIN fact_forecast_monthly AS ffm  
    ON fsm.date = ffm.date  
    AND fsm.customer_code = ffm.customer_code  
    AND fsm.product_code = ffm.product_code
```



UNION ALL

SELECT

ffm.date,

MONTHNAME(ffm.date) **AS** month_name, ffm.customer_code,

ffm.fiscal_year,

ffm.product_code, ffm.forecast_quantity, 0 **AS** sold_quantity

FROM fact_forecast_monthly **AS** ffm

LEFT JOIN fact_sales_monthly **AS** fsm

ON ffm.date = fsm.date

AND ffm.customer_code = fsm.customer_code

AND ffm.product_code = fsm.product_code

WHERE fsm.product_code **IS NULL**

)

SELECT

facr.month_name, facr.fiscal_year,

facr.customer_code, facr.product_code,

facr.forecast_quantity, facr.sold_quantity,

(forecast_quantity - sold_quantity) **AS** forecast_error

FROM forecast_actual_compare_report facr ;



Results

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```
fact_sales_monthly dim_customer fact_gross_price fact_forecast_monthly SQL File 11 x
WITH forecast_actual_compare_report AS (
    SELECT
        fsm.date,
        MONTHNAME(fsm.date) AS month_name , fsm.customer_code, fsm.product_code,
        fsm.fiscal_year,
        COALESCE(ffm.forecast_quantity, 0) AS forecast_quantity,
        fsm.sold_quantity
    FROM fact_sales_monthly AS fsm
    LEFT JOIN fact_forecast_monthly AS ffm
        ON fsm.date = ffm.date
        AND fsm.customer_code = ffm.customer_code
        AND fsm.product_code = ffm.product_code

    UNION ALL

    SELECT
        ffm.date,
        MONTHNAME(ffm.date) AS month_name, ffm.customer_code,
```

month_name	fiscal_year	customer_code	product_code	forecast_quantity	sold_quantity	forecast_error
September	2018	70002017	A0118150101	18	51	-33
September	2018	70002018	A0118150101	11	77	-66
September	2018	70003181	A0118150101	9	17	-8
September	2018	70003182	A0118150101	6	6	0

Result 1 x Read Only

Results

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fact_sales_monthlydim_customerfact_gross_pricefact_forecast_monthlySQL File 11* x

Limit to 50000 rows

14 UNION ALL

15

16 SELECT

17 ffm.date,

18 MONTHNAME(ffm.date) AS month_name, ffm.customer_code,

19 ffm.fiscal_year,

20 ffm.product_code, ffm.forecast_quantity, 0 AS sold_quantity

21 FROM fact_forecast_monthly AS ffm

22 LEFT JOIN fact_sales_monthly AS fsm

23 ON ffm.date = fsm.date

24 AND ffm.customer_code = fsm.customer_code

25 AND ffm.product_code = fsm.product_code

26 WHERE fsm.product_code IS NULL

27)

28 SELECT

29 facr.month_name, facr.fiscal_year,

30 facr.customer_code, facr.product_code,

31 facr.forecast_quantity, facr.sold_quantity,

32 (forecast_quantity - sold_quantity) AS forecast_error

33 FROM forecast_actual_compare_report facr ;

Result GridFilter Rows:Export:Wrap Cell Content:Fetch rows:

	month_name	fiscal_year	customer_code	product_code	forecast_quantity	sold_quantity	forecast_error
▶	September	2018	70002017	A0118150101	18	51	-33
	September	2018	70002018	A0118150101	11	77	-66
	September	2018	70003181	A0118150101	9	17	-8

Result 1 xRead Only

Results

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fact_sales_monthlydim_customerfact_gross_pricefact_forecast_monthlySQL File 11* x

Limit to 50000 rows

14 UNION ALL

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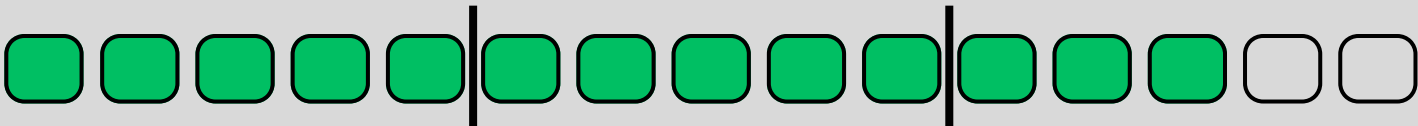
16 SELECT

17 ffm.date,

Result GridFilter Rows:Export:Wrap Cell Content:Fetch rows:

	month_name	fiscal_year	customer_code	product_code	forecast_quantity	sold_quantity	forecast_error
▶	September	2018	70002017	A0118150101	18	51	-33
	September	2018	70002018	A0118150101	11	77	-66
	September	2018	70003181	A0118150101	9	17	-8
	September	2018	70003182	A0118150101	6	6	0
	September	2018	70006157	A0118150101	5	5	0
	September	2018	70006158	A0118150101	6	7	-1
	September	2018	70007198	A0118150101	4	29	-25
	September	2018	70007199	A0118150101	7	34	-27
	September	2018	70008169	A0118150101	7	22	-15
	September	2018	70008170	A0118150101	8	5	3
	September	2018	70011193	A0118150101	5	10	-5
	September	2018	70011194	A0118150101	7	4	3
	September	2018	70012042	A0118150101	0	0	0
	September	2018	70012043	A0118150101	0	0	0
	September	2018	70013125	A0118150101	2	1	1
	September	2018	70013126	A0118150101	2	1	1
	September	2018	70016178	A0118150101	0	1	-1
	September	2018	70022085	A0118150101	12	20	-8

Result 1 xRead Only





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I Vishnu Ram Sachin I Data Analyst

SQL THROUGH ONE DATASET CHALLENGE